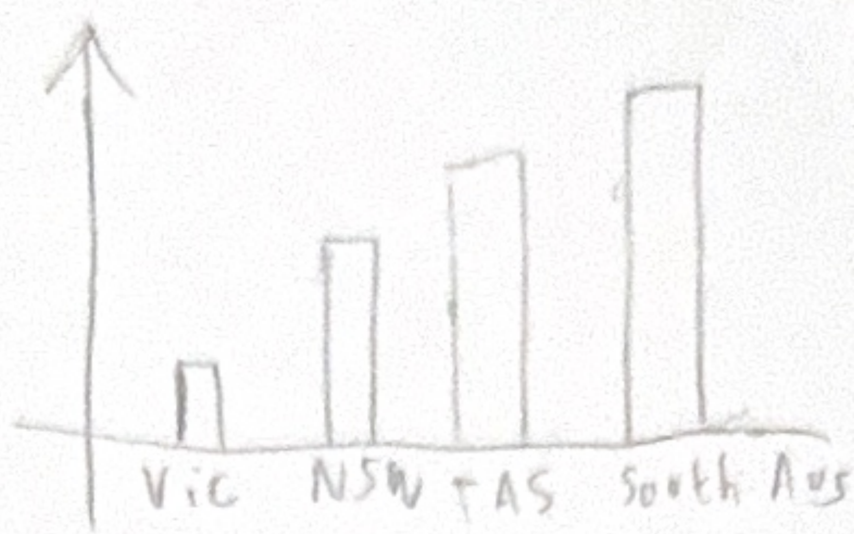
Title : Brainstorm
Author : Miguel Nicolas
Date : 05/10/2025
Sheet : 1
Task : Generate Design Ideas for topic

1) IDEAS

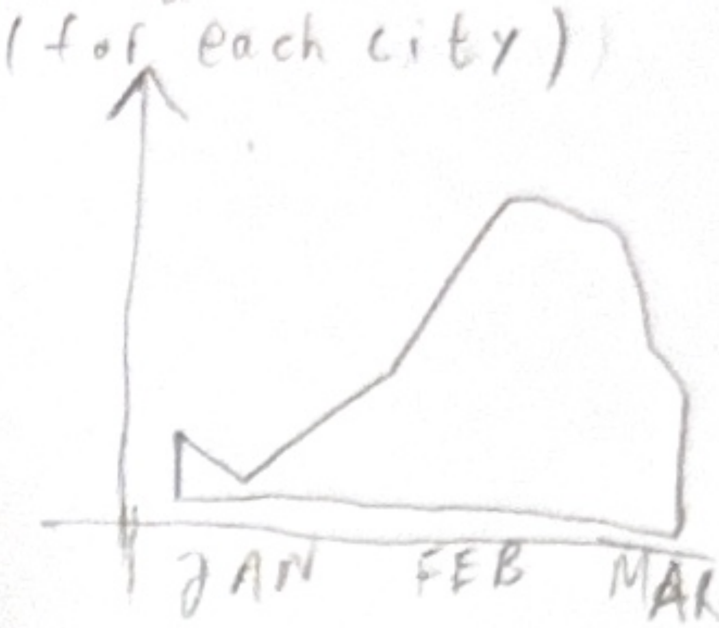
Australia lowest temperature



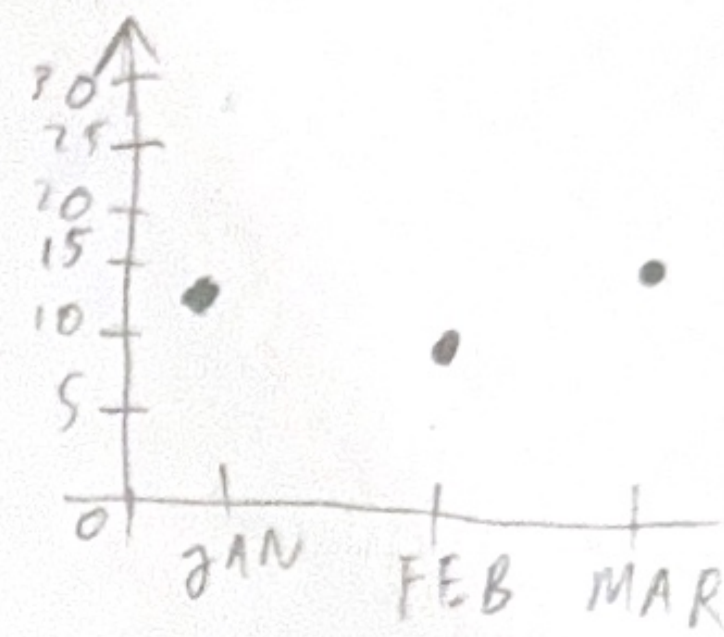
Extreme percentile trends

Total yearly rainfall

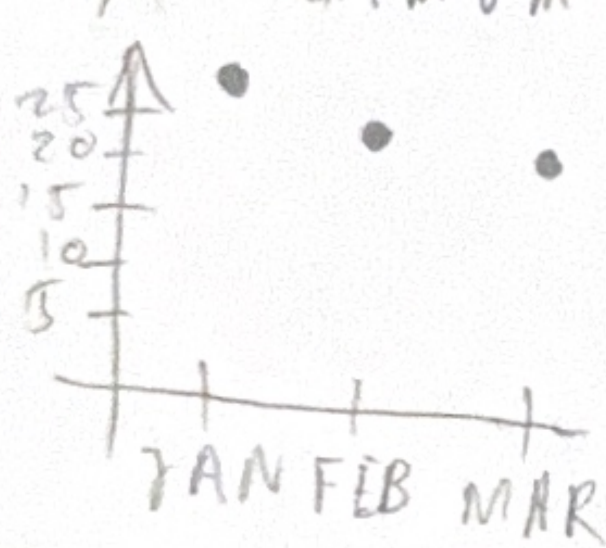
Annual Rainfall Australia (for each city)



City minimum temp

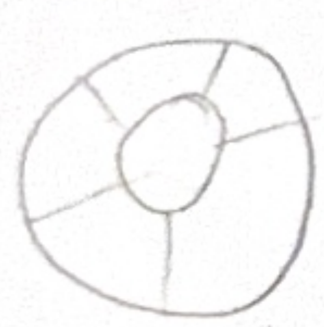


City Maximum Temp



2) FILTER

Donut chart

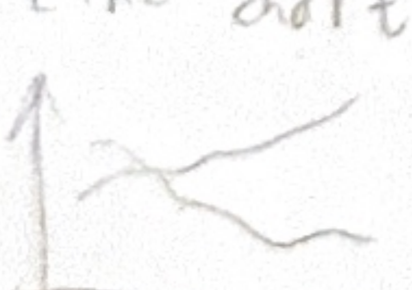


- Compare Australian cities by rainfall
- Letting a viewer quickly see patterns and then drill into one city.
- Hard to read when there's many cities/categories

Bar chart

- Easy to read
- Not ideal for trends, that need correlation

Line chart



- Excellent for showing trends over time
- Can become messy with many lines

Map

- Strong geografic context
- Need careful color choice

3) CATEGORISE

Climate zone by using the map

↓

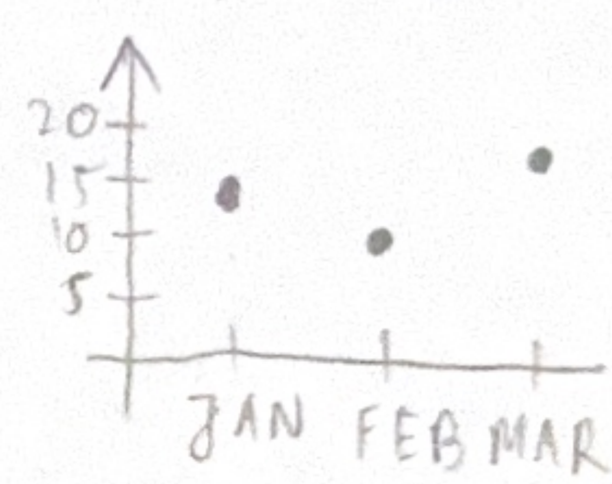
Show the yearly rainfall between each city

↓

Show the temperature for Australia's cities

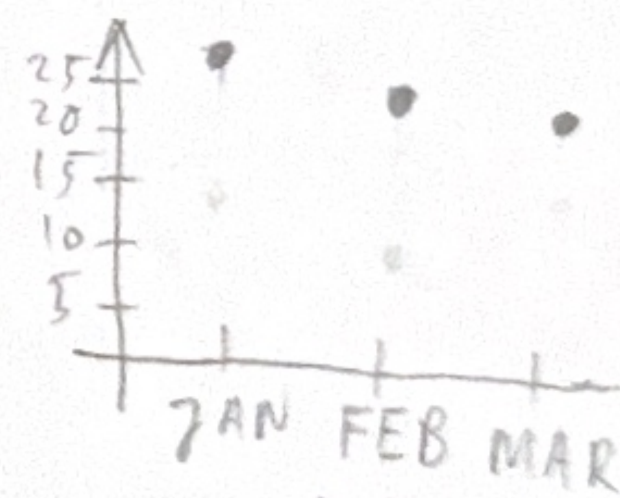
4) COMBINE & REFINE

City minimum temp



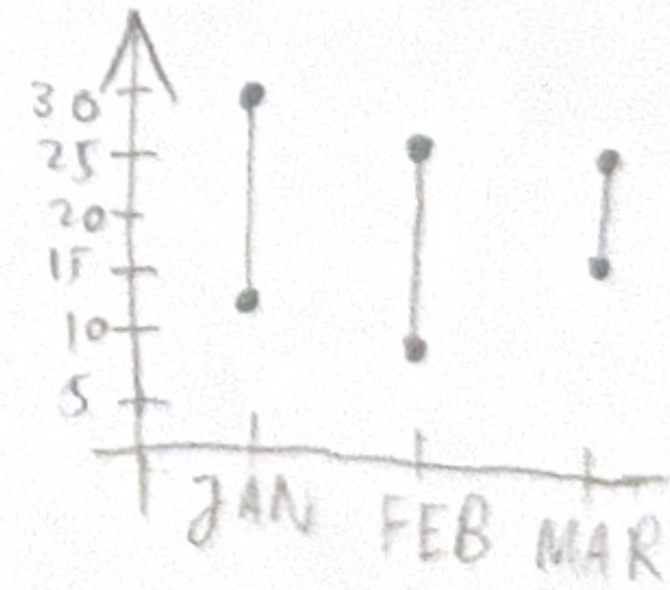
+

City maximum Temp



↓

City min & max Temp



5) QUESTIONS?

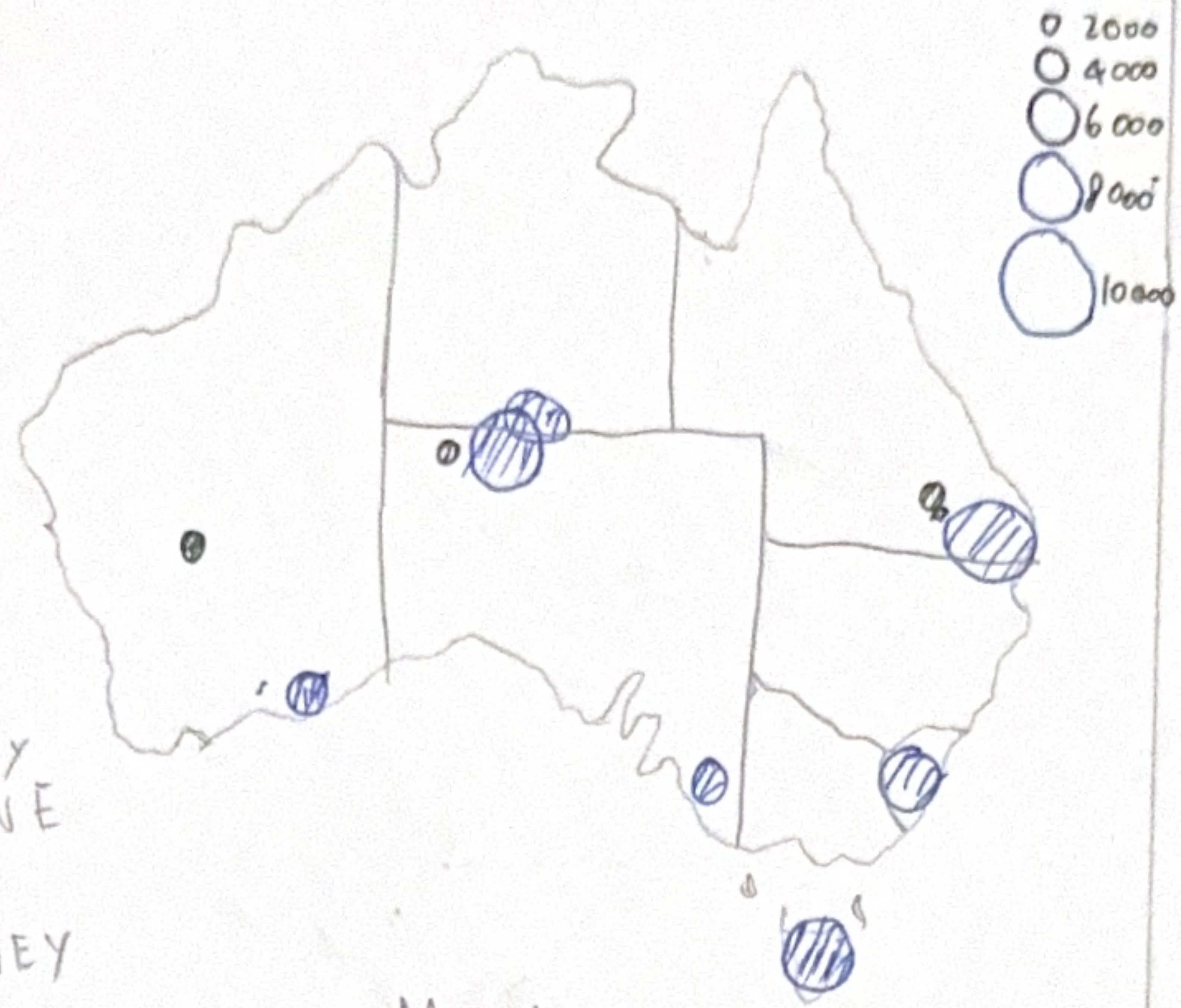
- Does the visualisation shows the true trend?
- What key insights or trends can help people understand Australia's climate diversity?
- How is this unique compare to other available visualisation?
- Does the design remain readable on different screen sizes or formats?

2) LAYOUT

AUSTRALIA WEATHER

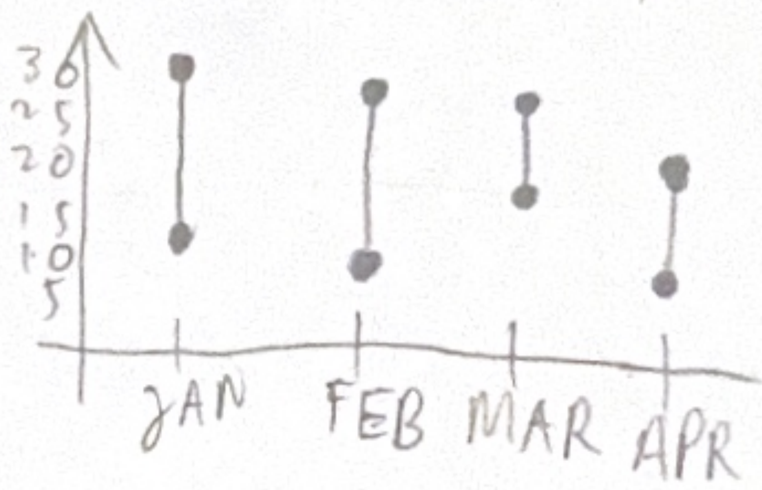
Text

City dots sized by rainfall



Total rainfall per city

Monthly min/max temp



Monthly Rainfall by city

- ☐ Adelaide
- ☐ Brisbane
- ☐ Canberra
- ☐ Darwin
- ☐ Sydney

Title : Idea 1
Author : Miguel Nicolas
Date : 05/10/2025
Sheet : 2
Task : Interactive Climate Dashboard

OPERATIONS

- Click on a city in the map → update donut chart.
- Hover on map dots → show city name + key stats
- Legend clearly explains colours, units, and data source
- Dropdown filter → changes the map colouring (by avg max temp, min temp)
- Zoom/pan on map → allow closer look at densely populated area

FOCUS

- Give users a clear overview of climate patterns across Australian cities
- Combine spatial (map) + temporal (line chart) + summary (donut) information
- Prioritise seasonal variation and total rainfall in a single clean interface

DISCUSSIONS

• Pros:

- Combines multiple perspectives (map + trend + summary) in one design
- Clear geographical context through the map
- Legend and filter allow easy interaction and user control

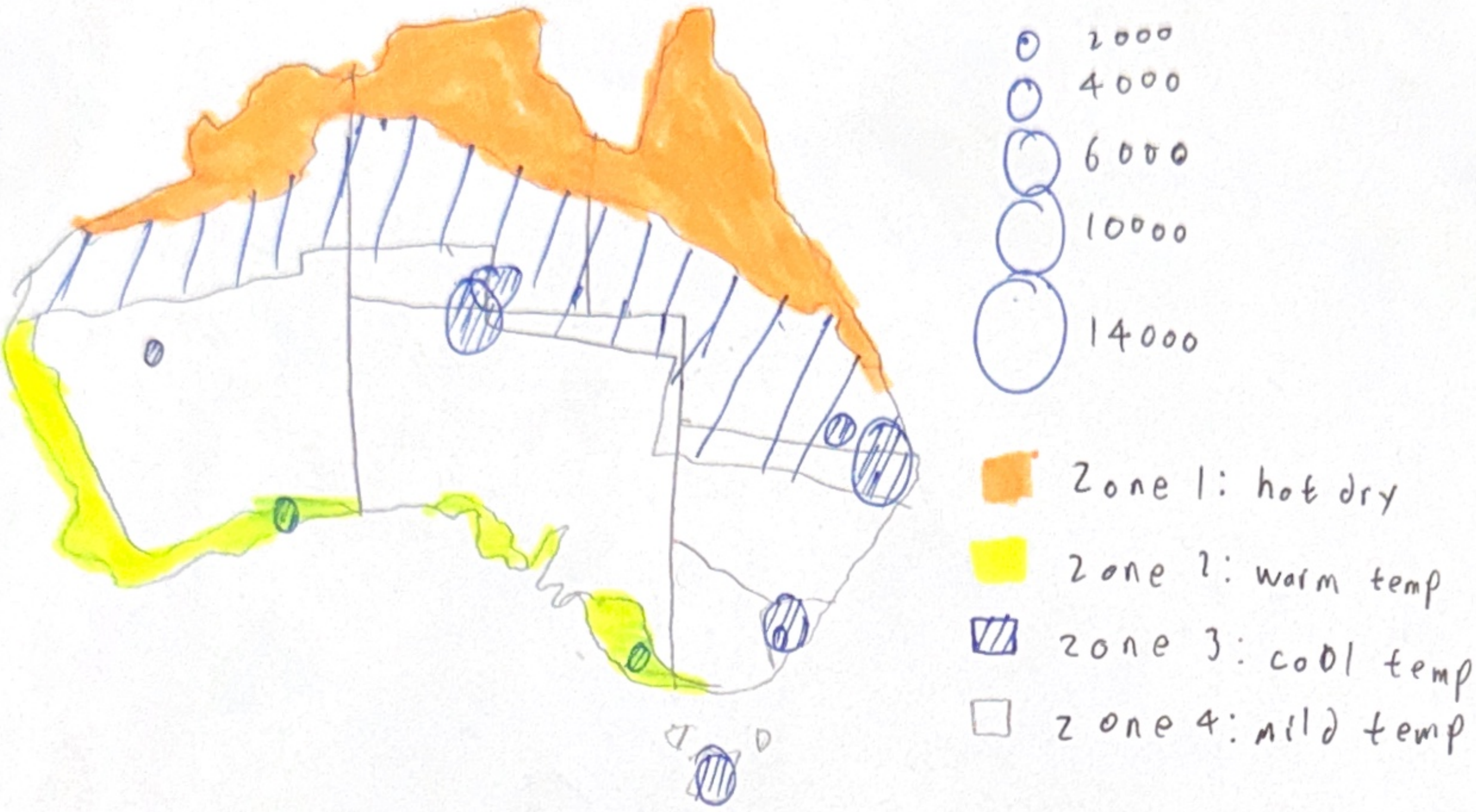
• Cons:

- Can look a bit crowded if too many cities are added
- Small screens may be harder to fit this layout nicely
- Too many interactions might overwhelm less experienced users.

3) LAYOUT

Title: Idea 2
Author: Miguel Nicolas
Date: 05/10/2025
Sheet: 3
Task: Australia's temp zones and extreme weather through map interaction

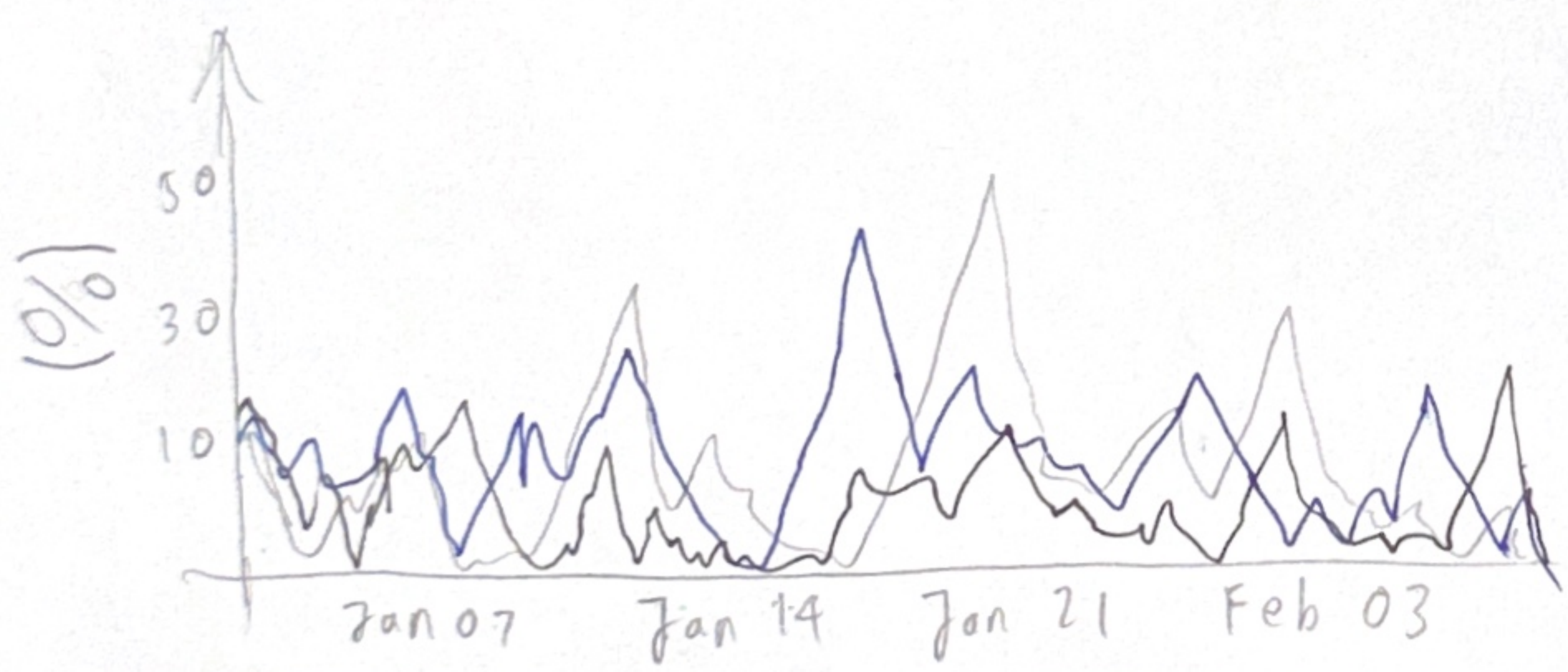
Exploring Temperature and Rainfall patterns



Text

Extreme weather categories

- Below 1st percentile
- Below 3rd percentile
- Above 97th percentile



OPERATIONS

- 1st percentile
  3rd percentile
  Above 97th

  Click to choose which categories want to focus
- Show/hide specific categories to compare patterns
- Toggle to return to the default full-map view

FOCUS

Show Australia's climate, which highlight regional climate differences

Communicate rainfall patterns across Australia

DISCUSSIONS

Pros:

- Showing zones by temperature provides clear, intuitive grouping
- Timeline charts effectively displays extreme weather over time

Cons:

- Timeline can get cluttered if many zones are selected
- Colouring for the temperature zones must be carefully chosen to avoid confusion

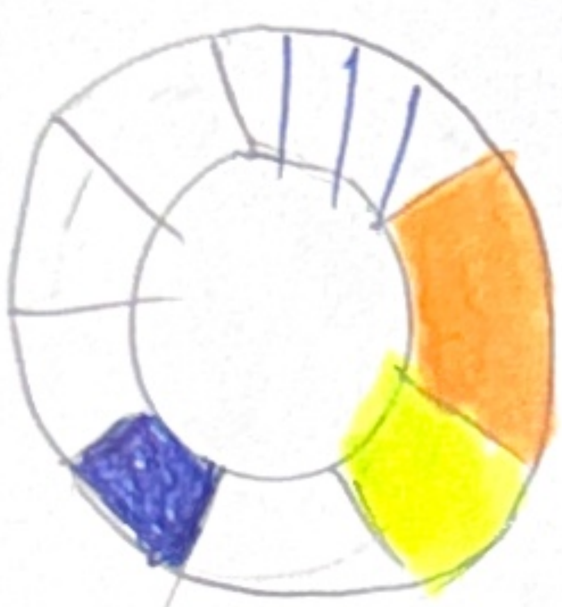
4) LAYOUT

AUSTRALIA CLIMATE & WEATHER

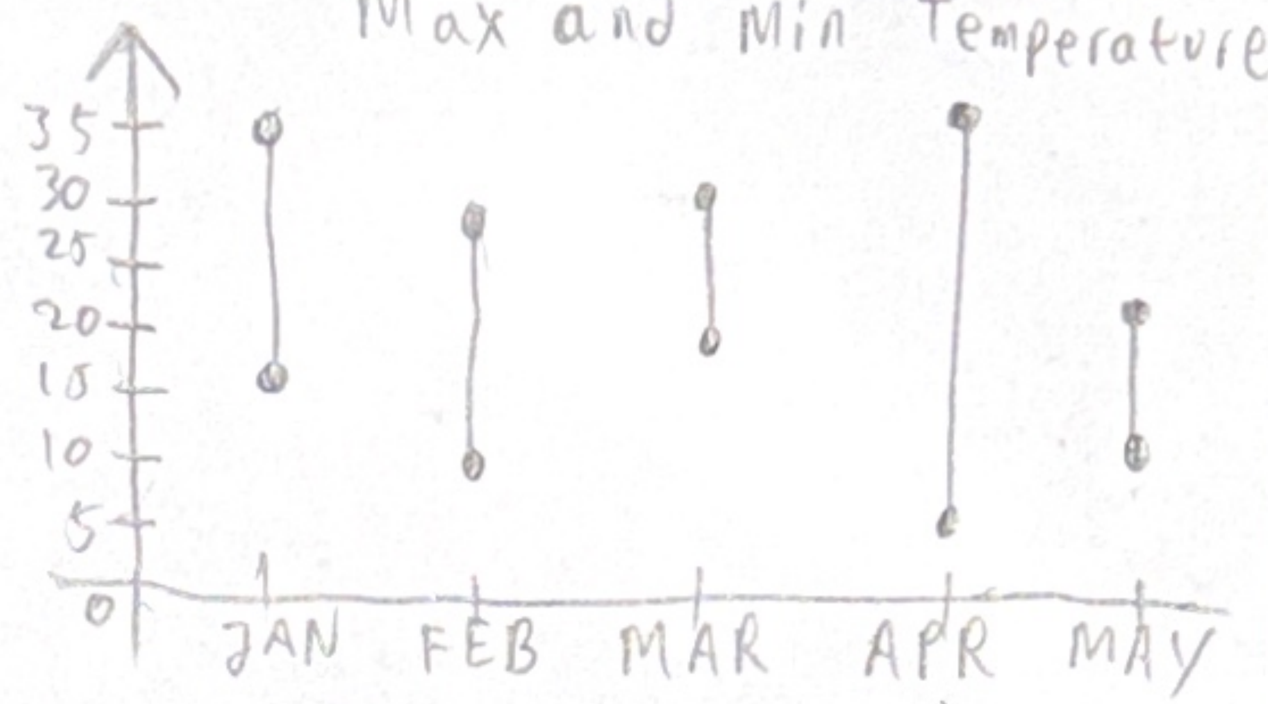
Text

- zone 1
- zone 2
- zone 3
- zone 4
- zone 5

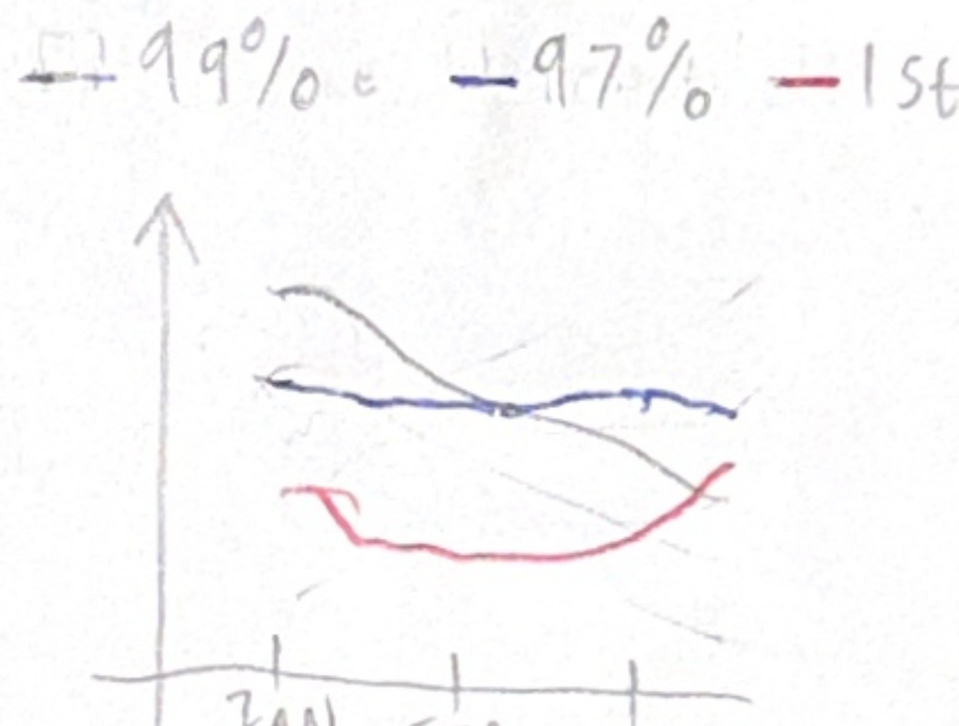
Sydney · Adelaide · Hobart · Perth



Annual Rainfall



Text



Extreme Percentile

Title: Idea 3
Author: Miguel Nicolas
Date: 05/10/2025
Sheet: 4
Task: Make climate zone map + City data dashboard

OPERATIONS

- zone 1
- zone 2
- zone 3
- zone 4
- zone 5

- Click on the zone filter to highlight the selected zone and a brief explanation on the left will display

Sydney · Adelaide · Hobart · Perth

- User can filter and choose which specific city want to choose on min/max temp

DISCUSSIONS

Pros:

- Simple and interactive easy to see the direct comparison
- users can choose which specific category to highlight

Cons:

- Needs careful color and spacing design
- Mainly focus on cities only

FOCUS

- To help users understand how Australia's climate varies spatially and temporally
- Make official data more accesible and engaging through interactive and consistent visual storytelling

5) LAYOUT

Title: Final design (Realization)

Author: Miguel Nicolos

Date: 06/10/2025

Sheet: 5

Task: Build Australia's Climate varies by region and time

zone 1

zone 2

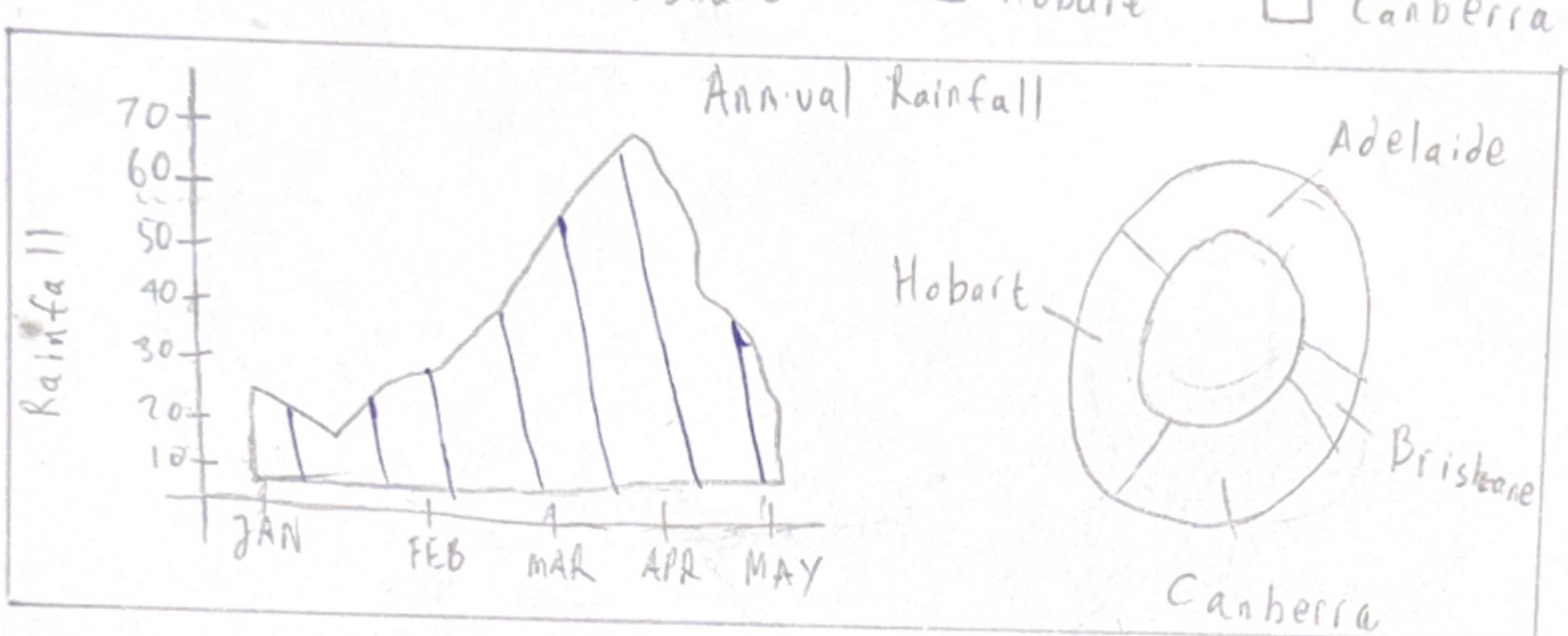
zone 3

zone 4

zone 5

} Text

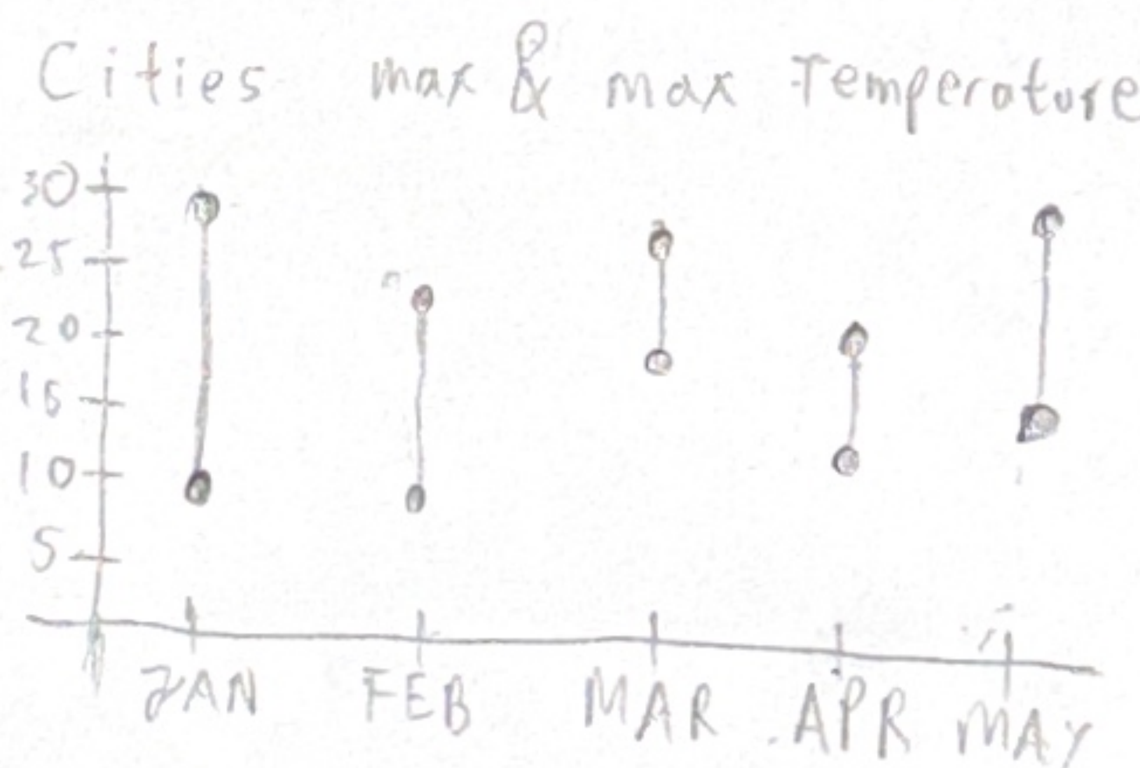
☐ Adelaide ☐ Brisbane ☐ Hobart ☐ Canberra

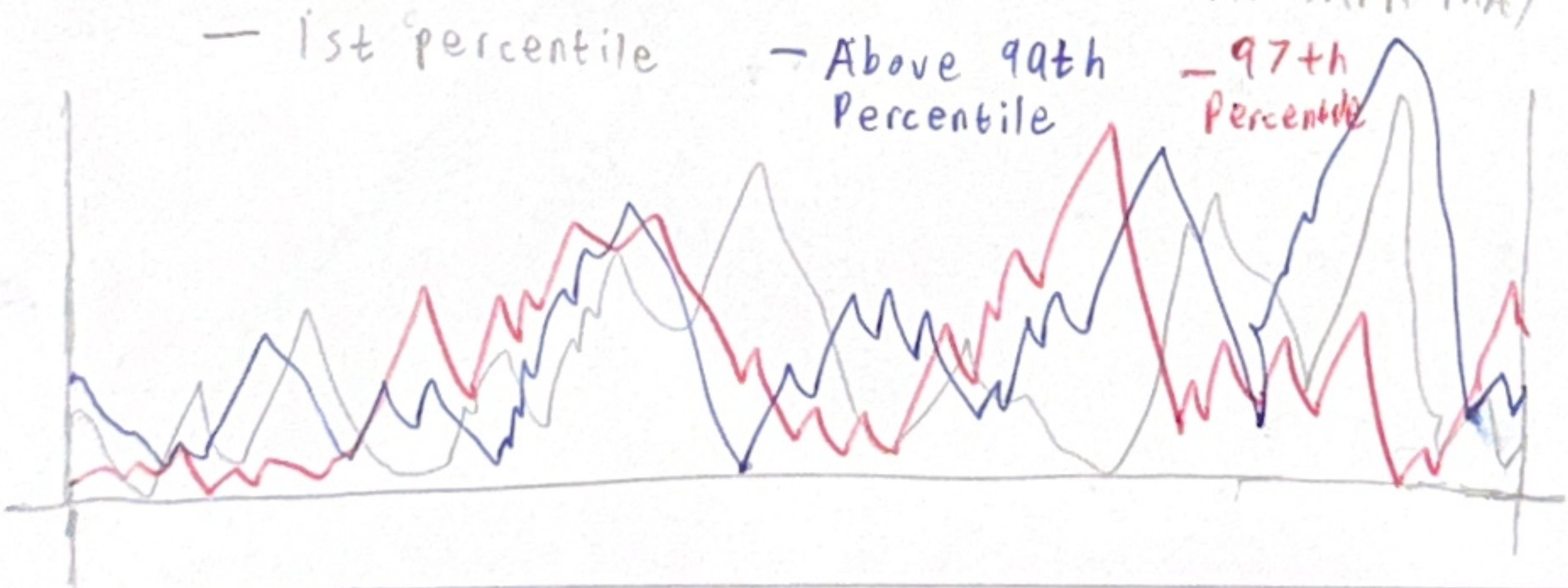


Choose city

Adeloide ▽

⇒





OPERATIONS

☐ Adelaide ☐ Brisbane

- Clicking a city changes both the rainfall donut and area charts to show that city's data

Choose city

Sydney ▽

- Choosing a city from the dropdown updates the min-max temperature chart

DETAILS

- Dependencies
  - Vega-Lite
  - Phyton
  - HTML + CSS
- Algorithm
  - Implement the system to toggle between categories
- Requirements
  - Browser that supports CSS 3 and HTML 5
  - Screen resolution with at least 480 x 320 resolution

Focus

- The climate zones show how they relate to cities temperature and rainfall patterns
- Communicate how "extreme" conditions differ between cities